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SCHOOL / DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB PROJECT REPORT

ON

YOUTUBE VIDEO STATISTICS ANALYSIS USING PYSPARK

Course / Lab: Big Data Analytics

Course Code: 24CS02PR0503

Submitted By

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Academic Year: 2026–27

CERTIFICATE

This is to certify that the project entitled “YouTube Video Statistics Analysis using PySpark” has been successfully completed by Arpita Hanwatkar, Roll No. 04, and Maithili Mahajan, Roll No. 12, of the Computer Science and Engineering (Artificial Intelligence and Data Science) programme, Ramdeobaba University, Nagpur, as part of the requirements of the laboratory course Big Data Analytics during the academic year 2026–27.

The work presented in this report has been carried out under the guidance and supervision of the undersigned.

DECLARATION

We hereby declare that the project work entitled “YouTube Video Statistics Analysis using PySpark” is our original work carried out for the laboratory course at Ramdeobaba University, Nagpur. The work has not been submitted elsewhere for the award of any academic qualification.

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Date: 27/10/2026 Place: Nagpur

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ABSTRACT

YouTube generates large volumes of video statistics, including views, likes, dislikes, comments, channel information, and category identifiers. Examining these records helps explain video popularity, channel performance, and audience engagement. This project presents an analysis of YouTube Trending Video Statistics using Apache Spark through PySpark. The dataset initially contained 37,352 records and 16 columns. Duplicate removal reduced the data to 33,320 rows, and cleaning selected numeric fields produced a final analysis dataset of 33,088 rows. The workflow includes data loading, duplicate removal, missing-value handling, numeric type conversion, exploratory data analysis, aggregation, ranking, engagement-rate calculation, and visualization. PySpark DataFrames and group-by aggregations were used to examine video, channel, and category performance. The engagement rate was calculated as ((likes + comment_count) / views) × 100 for records with positive views. The results showed that T-Series had the highest total views among the analyzed channels, while category ID 24 had the highest total views among the analyzed categories. Scatter plots showed an overall positive relationship between views and likes, and between views and comments. The notebook also prepared dashboard KPIs and visual outputs for exploring the findings. The project demonstrates how PySpark can support structured analysis of video statistics and turn raw records into interpretable insights.

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1. INTRODUCTION

1.1 Background

Online video platforms generate substantial data through viewing activity and audience interactions. Each record in a trending-video dataset can contain video metadata, channel name, category, views, likes, dislikes, comments, publication time, and other descriptive fields. Reviewing such data manually is difficult, particularly when comparisons across thousands of records are required. Apache Spark provides a distributed data-processing framework, and PySpark exposes Spark functionality through Python. This project uses PySpark DataFrames and aggregation operations to organize and analyze YouTube video statistics.

1.2 Motivation

The project was undertaken to understand how engagement indicators vary across videos, channels, and categories. It also provides practical experience with big-data processing concepts, data cleaning, aggregation, exploratory analysis, and visualization in a notebook-based environment.

1.3 Problem Definition

The problem is to process a large collection of YouTube trending-video records and extract meaningful information about video reach and audience interaction. The analysis must address duplicate records, missing values in important fields, and numeric values that require conversion before calculations. It must then summarize and visualize video, channel, category, and engagement patterns.

1.4 Scope

The project covers loading the INvideos.csv dataset, inspecting its structure, removing duplicates, handling missing values in important columns, converting selected metrics to numeric types, ranking videos, aggregating statistics by channel and category, calculating engagement rate, and producing plots and dashboard KPI values. It is an offline analysis of the supplied dataset; live YouTube API collection and predictive machine learning are not part of the implemented workflow.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

To analyze YouTube Trending Video Statistics efficiently using PySpark by cleaning the dataset, identifying high-performing videos, comparing channels and categories, examining the relationship between views and audience interactions, and presenting the results through clear visualizations and dashboard summaries.

2.2 Objectives

1. To clean and preprocess the YouTube Trending Video Statistics dataset by handling duplicate records, missing values, and converting numerical columns into appropriate data types.

2. To identify the most popular YouTube videos based on views, likes, and comments, and to compare the performance of different channels and video categories.

3. To calculate engagement rates and analyze the relationships between video views, likes, and comments using PySpark and data visualizations.

4. To develop an interactive dashboard using Gradio and Plotly to present video rankings, channel performance, category analysis, and engagement results in an understandable format.

2.3 Expected Outcome

The expected outcome is a cleaned dataset and a set of ranked tables, aggregate summaries, engagement-rate results, and plots that help interpret YouTube video performance. The completed analysis should make it easier to compare channels and categories and to observe broad relationships between views and interactions.

3. REQUIREMENTS

3.1 Hardware Requirements

• Computer or laptop capable of running a modern web browser.\n• Internet connection for Google Colab and access to the dataset.\n• Local hardware requirements depend on the browser and workload; the project was developed in Google Colab.

3.2 Software Requirements

• Platform: Google Colab notebook environment.\n• Language: Python.\n• Processing framework: Apache Spark with PySpark (Spark version reported as 4.0.4 in the notebook).\n• Data handling: Pandas for converting limited result sets for visualization.\n• Visualization: Matplotlib; Plotly and ipywidgets were installed/imported for dashboard-related work.\n• Input format: CSV (INvideos.csv).

3.3 Dataset / Input Requirements

The project uses the YouTube Trending Video Statistics dataset loaded from INvideos.csv. The source dataset contained 37,352 rows and 16 columns, including video_id, trending_date, title, channel_title, category_id, publish_time, tags, views, likes, dislikes, comment_count, thumbnail_link, comments_disabled, ratings_disabled, video_error_or_removed, and description. The description field had 561 missing values in the initial pandas inspection. The notebook's Spark read reported 38,528 rows before duplicate removal; after dropDuplicates(), 33,320 rows remained. Removing records with missing values in important fields left 33,088 rows for the analysis. Numeric metrics were then cast to BIGINT and invalid/missing metric rows were dropped before analysis.

4. METHODOLOGY / ALGORITHM

4.1 Proposed Method

The proposed method focuses on analyzing YouTube Trending Video Statistics using Apache Spark (PySpark). The dataset is loaded from a CSV file, and its schema, number of records, column names, and missing values are inspected. Duplicate records are removed, and rows containing missing values in essential fields are excluded.

The required columns are then selected, and numerical attributes such as views, likes, dislikes, and comment count are safely converted into appropriate numeric data types. Records containing invalid numerical values are removed to ensure reliable analysis.

The cleaned dataset is used to identify top-performing videos based on views, likes, and comments. Group-by operations are performed to calculate the total views, likes, and comments for individual channels and video categories. An engagement rate is also calculated using likes, comments, and views for records with positive view counts.

Finally, the analysis results are converted into limited Pandas DataFrames for visualization. Bar charts, scatter plots, ranked tables, and KPI summaries are prepared using Plotly and other plotting tools. An interactive dashboard developed using Gradio allows users to explore video rankings, channel performance, category analysis, and engagement results.

4.2 Algorithm

Step 1: Initialize the SparkSession and load the YouTube Trending Video Statistics dataset from the CSV file.

Step 2: Inspect the dataset schema, record count, column names, and missing values. Remove duplicate records and rows with missing values in essential fields.

Step 3: Select the required columns, safely convert views, likes, dislikes, and comment count into numeric data types, and remove records containing invalid numerical values.

Step 4: Rank videos based on views, likes, and comments. Aggregate views, likes, and comments by channel and category, and calculate the engagement rate for records with positive views.

Step 5: Prepare summary tables, KPI metrics, and visualizations. Display the analysis results through an interactive Gradio and Plotly dashboard.

4.3 Flowchart

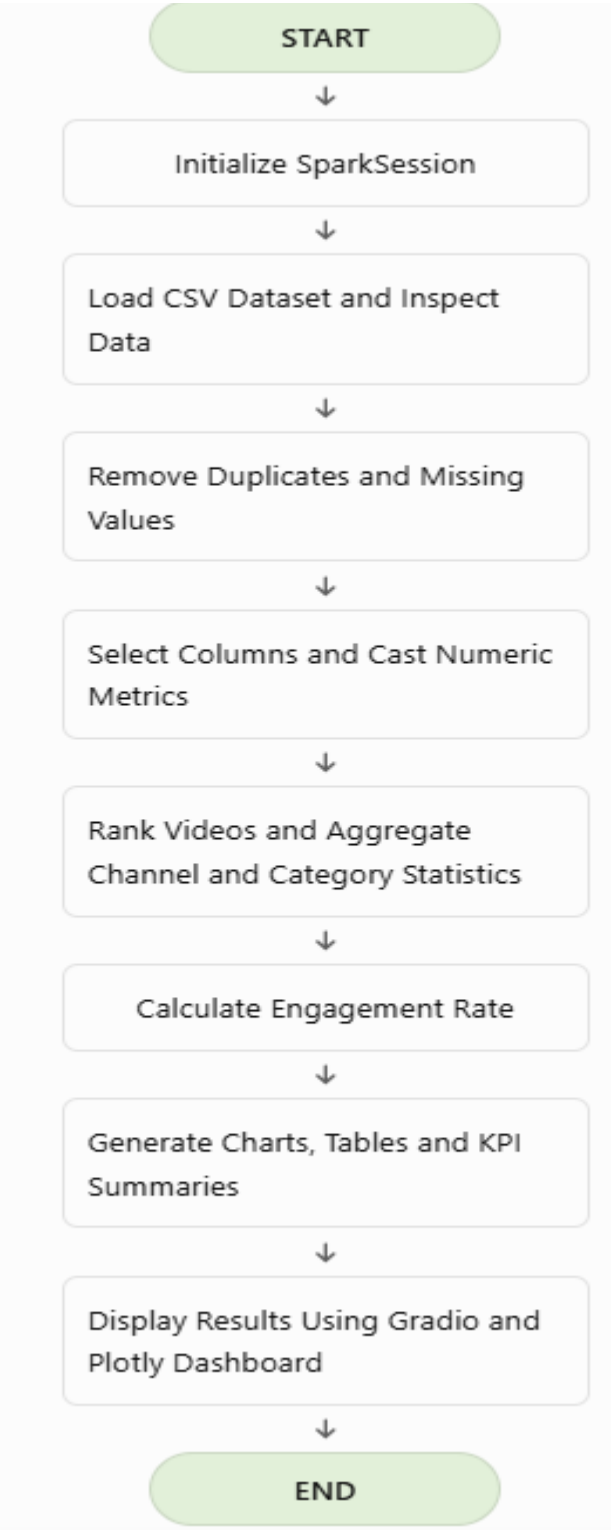


Figure 4.1: Flowchart of the proposed YouTube Video Statistics Analysis methodology.

4.4 Pseudocode

```
BEGIN
    Create SparkSession
    Read YouTube CSV dataset
    Inspect schema, columns, record count and missing values
    Remove duplicate records
    Remove rows missing essential fields
    Select required analysis columns
    Safely convert numerical metrics into numeric types
    Remove records with invalid numerical metrics
    Rank videos by views, likes and comments
    Group records by channel
    Calculate total views, likes and comments
    Group records by category
    Calculate total views, likes and comments
    FOR each record with views > 0
        Calculate engagement_rate =
            ((likes + comment_count) / views) * 100
    END FOR

    Prepare summary tables and KPI metrics
    Convert limited results into Pandas DataFrames
    Generate charts and visualizations
    Display results using Gradio and Plotly

END
```

5. SYSTEM DESIGN

5.1 System Architecture

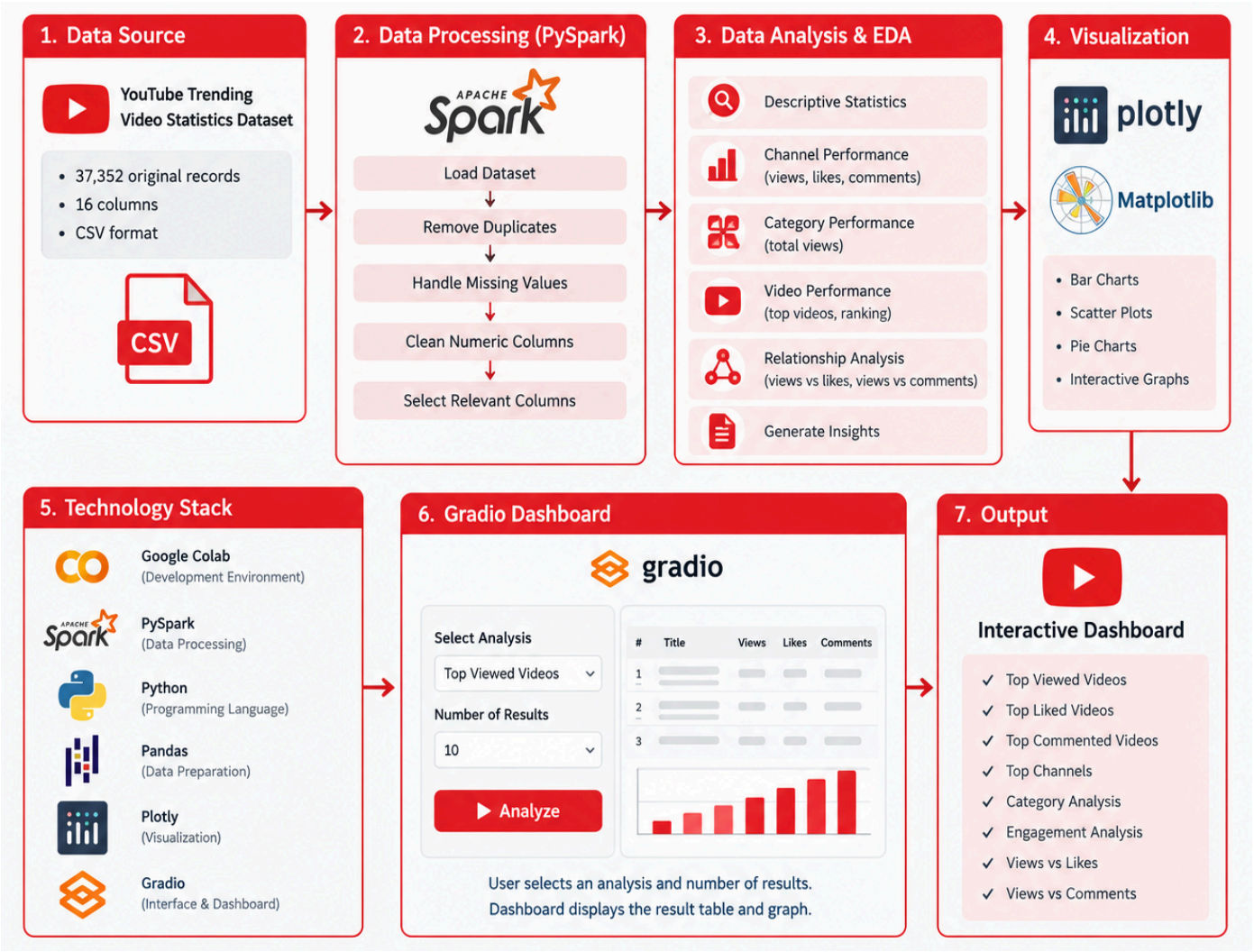


Figure : System Architecture

5.2 Modules

The project is divided into the following modules to perform YouTube Trending Video Statistics analysis systematically.

1. Data Loading and Inspection Module

This module loads the YouTube dataset from a CSV file into a PySpark DataFrame. It examines the dataset dimensions, column names, schema, record count, and missing values to understand the structure and quality of the data.

2. Data Cleaning Module

This module removes duplicate records and eliminates rows containing missing values in essential fields, such as video ID, title, channel name, category ID, and engagement metrics. It prepares the dataset for further analysis.

3. Data Type Conversion Module

This module selects the required columns and safely converts numerical attributes, including views, likes, dislikes, and comment count, into BIGINT data types. Records containing invalid numerical values are removed to maintain data consistency.

4. Video Analysis Module

This module identifies high-performing videos by sorting and ranking records according to views, likes, and comments. It helps determine which videos receive higher visibility and audience interaction.

5. Channel and Category Analysis Module

This module groups the cleaned dataset by channel name and category ID. It calculates the total views, likes, and comments for each group, enabling comparisons of channel performance and category popularity.

6. Engagement Analysis Module

This module calculates the engagement rate using the number of likes, comments, and views for records with positive views. The results are sorted to identify the top ten records with higher engagement rates.

The engagement rate is calculated using the following formula:

$$EngagementRate = \frac{Likes+Comments}{Views} \times 100$$

7. Visualization and Dashboard Preparation Module

This module prepares limited result sets as Pandas DataFrames for plotting and visualization. It generates charts, ranked tables, and Key Performance Indicator (KPI) summaries. The results are presented through an interactive dashboard developed using Gradio and Plotly.

5.3 Data Flow / Workflow

The data flow describes how the dataset moves through different processing and analysis stages in the proposed system.

Initially, the YouTube Trending Video Statistics CSV file is loaded into a PySpark DataFrame. The dataset is inspected to examine its structure, schema, record count, and missing values. Data cleaning operations remove duplicate records and rows with missing values in essential fields.

After cleaning, the required columns are selected, and numerical metrics are converted into suitable data types. Invalid numerical records are removed to prepare a consistent dataset for analysis.

The cleaned DataFrame is then used for four major analytical operations:

1. Video-level analysis: Ranks videos according to views, likes, and comments.
2. Channel-level analysis: Calculates total views, likes, and comments for each channel.
3. Category-level analysis: Calculates total views, likes, and comments for each video category.
4. Engagement analysis: Calculates engagement rates and identifies records with high engagement.

Finally, selected results are converted into Pandas DataFrames for plotting, while overall metrics are collected to calculate dashboard KPIs, such as the number of analyzed records, total views, total likes, and total comments. Charts and summary tables are then presented through the Gradio and Plotly dashboard for convenient exploration and interpretation.

Data Flow Diagram



Figure : Data Flow Diagram

6. IMPLEMENTATION

6.1 Development Environment

The project, YouTube Video Statistics Analysis using Apache Spark (PySpark), was developed using Google Colab, Python, and PySpark. Google Colab provided a notebook-based environment for executing code, processing the dataset, and performing data analysis.

A SparkSession named YouTube Video Statistics Analysis was created to initialize the Spark environment. The notebook reported Apache Spark version 4.0.4. The dataset, INvideos.csv, was loaded using Spark's CSV reader with the header option enabled and schema inference applied.

PySpark DataFrames were used for data cleaning, numerical type conversion, sorting, grouping, and aggregation. Pandas was used to convert selected result tables into a format suitable for visualization. Matplotlib was used to generate bar charts and scatter plots. Plotly and ipywidgets were also imported or installed for dashboard-related exploration, while Gradio was used for the interactive dashboard interface.

Tools and technologies used:

- Programming Language: Python
- Development Environment: Google Colab
- Big Data Processing: Apache Spark (PySpark)
- Data Handling: PySpark DataFrames and Pandas
- Data Visualization: Matplotlib and Plotly
- Dashboard Interface: Gradio
- Dataset Format: CSV (INvideos.csv)

6.2 Module-wise Implementation

1. Data Loading and Inspection

The dataset is loaded into a PySpark DataFrame using the CSV reader. The schema, column names, number of records, dataset dimensions, and missing values are inspected to understand the structure and quality of the input data.

2. Data Cleaning

The dropDuplicates() function removes duplicate records from the dataset. The dropna() function is applied to essential fields to exclude records containing missing values that could affect the analysis.

3. Data Type Conversion

The required columns are selected for analysis. Numerical attributes such as views, likes, dislikes, and comment_count are safely converted into BIGINT values using try_cast. Records containing invalid numerical values are removed to maintain data consistency.

4. Video Analysis

The orderBy() function is used to sort video records according to views, likes, and comments in descending order. This helps identify the videos with the highest recorded audience reach and interactions.

5. Channel and Category Analysis

The groupBy() and aggregation functions calculate the total views, likes, and comments for each channel and category. The aggregated results are sorted to identify channels and categories with higher total views.

6. Engagement Analysis

An engagement_rate column is created using likes, comments, and views. For records with positive views, the engagement rate is calculated as a percentage. For records with zero or negative views, the value is set to zero using the otherwise() condition.

7. Visualization and Dashboard Preparation

The top-ten result tables and selected summary metrics are prepared for visualization. Limited result sets are converted into Pandas DataFrames for plotting. Bar charts, scatter plots, and summary metrics are used to present the findings, which can be explored through the dashboard interface.

6.3 Important Code Snippets

1. Creating a SparkSession and Loading the Dataset

This code initializes the Spark environment and loads the CSV dataset with column headers and automatic schema inference.

```
None
from pyspark.sql import SparkSessionspark = SparkSession.builder \ .appName("YouTube Video Statistics Analysis") \ .getOrCreate()df = spark.read.csv( "INvideos.csv", header=True, inferSchema=True)
```

2. Channel-wise Aggregation

This code groups the dataset by channel name and calculates the total views, likes, and comments for each channel. The results are sorted by total views in descending order.

```
None
from pyspark.sql.functions import sum, colchannel_analysis = youtube_df.groupBy("channel_title").agg( sum("views").alias("total_views"), sum("likes").alias("total_likes"), sum("comment_count").alias("total_comments")).orderBy(col("total_views").desc())channel_analysis.show(10, truncate=False)
```

3. Calculating the Engagement Rate

This code creates an engagement_rate column using the number of likes, comments, and views. It calculates the percentage only when views are greater than zero; otherwise, the engagement rate is set to zero.

```
None
from pyspark.sql.functions import col, whenyoutube_df = youtube_df.withColumn( "engagement_rate", when( col("views") > 0, ( (col("likes") + col("comment_count")) / col("views") ) * 100 ).otherwise(0))
```

The formula used is:

$$EngagementRate = \frac{Likes+Comments}{Views} \times 100$$

4. Category-wise Aggregation

This code groups video records by category ID and calculates the total views, likes, and comments for each category.

```
None
category_analysis = youtube_df.groupBy("category_id").agg( sum("views").alias("total_views"), sum("likes").alias("total_likes"), sum("comment_count").alias("total_comments")).orderBy(col("total_views").desc())category_analysis.show(10)
```

5. Ranking Videos by Views

This code sorts video records according to their view counts in descending order and displays the top ten records.

```
None
top_videos = youtube_df.orderBy( col("views").desc())top_videos.select( "title", "channel_title", "views").show(10, truncate=False)
```

7. RESULTS AND DISCUSSION

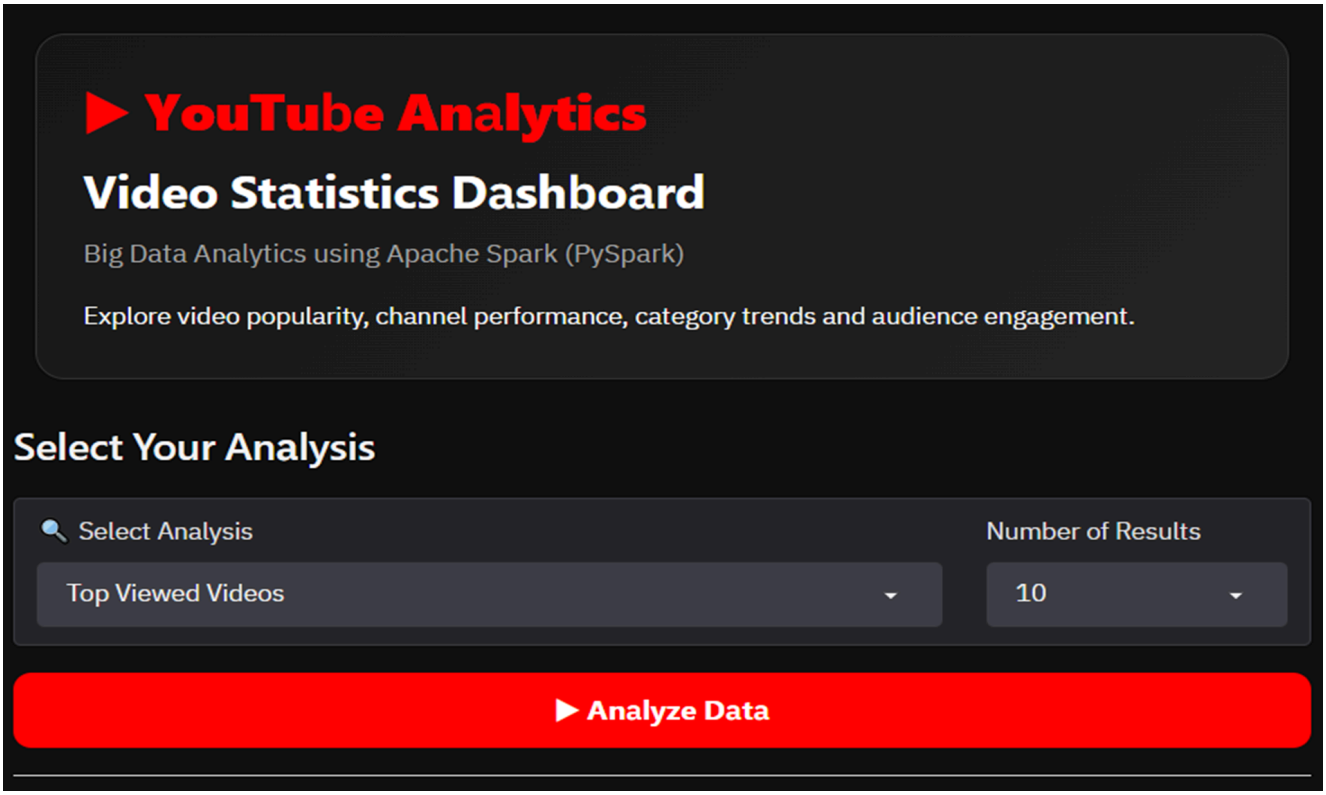
7.1 Experimental Setup

The YouTube Trending Video Statistics analysis was performed using Google Colab, Python, and PySpark. The dataset, INvideos.csv, was inspected initially using Pandas, which reported 37,352 rows and 16 columns. When the dataset was loaded using the Spark CSV reader, 38,528 rows were reported before duplicate removal.

The dropDuplicates() operation reduced the dataset to 33,320 records. Further cleaning, including the removal of records with missing essential fields and invalid numerical metrics, resulted in a final analysis dataset containing 33,088 valid records. The columns views, likes, dislikes, and comment_count were converted to BIGINT data types.

The cleaned dataset was used to perform video rankings, channel-wise and category-wise aggregations, engagement-rate calculations, and visualization. The channel and category rankings were sorted in descending order of total views. Summary metrics were also prepared for the dashboard.

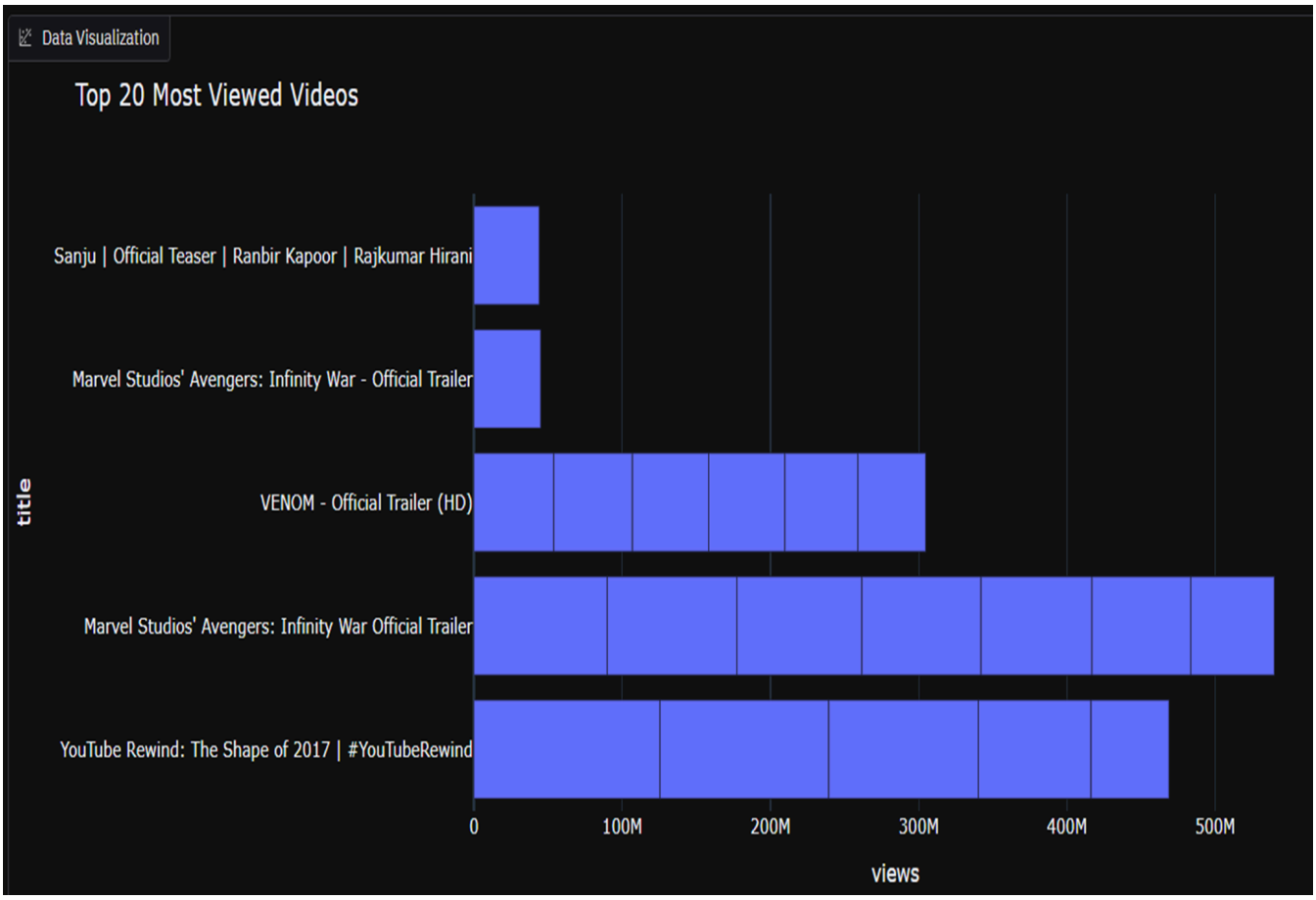
7.2 Results



Analysis Results

Results Table

title	channel_title	views	likes	comment_count
YouTube Rewind: The Shape of 2017 | #YouTubeRewind	YouTube Spotlight	125432237	2912710	807558
YouTube Rewind: The Shape of 2017 | #YouTubeRewind	YouTube Spotlight	113876217	2811216	787174
YouTube Rewind: The Shape of 2017 | #YouTubeRewind	YouTube Spotlight	100911567	2656672	682890
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	89930713	2606663	347982
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	87449453	2584674	341571
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	84281319	2555411	339708
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	80360459	2513102	335920
YouTube Rewind: The Shape of 2017 | #YouTubeRewind	YouTube Spotlight	75969469	2251815	827755
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	74789251	2444956	330710
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	66637636	2331352	316185
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	56367282	2157733	303178
VENOM - Official Trailer (HD)	Sony Pictures Entertainment	53822757	1250535	135405



7.3 Performance Analysis

The project evaluates analytical results through data-cleaning counts, aggregate statistics, ranked outputs, and visualization patterns rather than predictive-model accuracy.

The following table summarizes the key findings.

Analysis	**Result**
Highest-viewed channel by aggregate views	T-Series
Total views for T-Series	1,748,057,724
Second-highest channel by aggregate views	Marvel Entertainment
Total views for Marvel Entertainment	1,058,174,340
Third-highest channel by aggregate views	FoxStarHindi
Total views for FoxStarHindi	982,953,616
Highest-viewed category	Category ID 24
Total views for category ID 24	13,326,399,551
Second-highest category	Category ID 10
Total views for category ID 10	8,336,504,205
Highest observed engagement rate	Approximately 77.43%

The engagement rate is a descriptive measure calculated from likes and comments relative to views. It helps identify records with high audience interaction in relation to their view counts.

7.4 Discussion

The results demonstrate that total view counts vary considerably across channels and categories. T-Series ranked first among the analyzed channels, while category ID 24 recorded the highest aggregate views. These findings illustrate how PySpark aggregation operations can be used to compare channel reach and category-level popularity.

The scatter plots of views versus likes and views versus comments showed an overall positive relationship. This suggests that videos with higher view counts often also receive more likes and comments. However, this overall pattern does not establish that views directly cause higher engagement.

Some technology-related and giveaway videos recorded particularly high engagement rates. Such results highlight the usefulness of engagement analysis for identifying videos that receive substantial audience interaction relative to their views. Nevertheless, engagement rate alone does not measure audience satisfaction, content quality, or long-term popularity.

Overall, the project demonstrates how PySpark can support large-scale data cleaning, aggregation, ranking, and exploratory visualization. The resulting tables and dashboard summaries make the dataset easier to interpret and compare.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project, YouTube Video Statistics Analysis using Apache Spark (PySpark), demonstrates an end-to-end workflow for processing and analyzing YouTube Trending Video Statistics. The dataset was inspected, duplicate records were removed, essential fields were cleaned, and numerical attributes were converted into suitable data types.

The final cleaned analysis dataset contained 33,088 records. PySpark operations were used to rank videos by views, likes, and comments and to aggregate statistics by channel and category. Engagement rates were also calculated to examine audience interactions relative to video views.

The results identified T-Series as the channel with the highest aggregate views among the analyzed channels and category ID 24 as the category with the highest total views. Visualizations showed overall positive relationships between views and likes and between views and comments.

The project also prepared KPI summaries and an interactive dashboard using Gradio and Plotly to present the analytical results. Overall, the work demonstrates the usefulness of PySpark for data processing and exploratory analytics and provides a structured approach to understanding video popularity, channel performance, category trends, and engagement.

8.2 Future Scope

The project can be extended in the following ways:

1. Advanced Filtering: Add filters for trending dates, video categories, and channel names to enable more detailed analysis.

2. Real-Time Data Integration: Integrate the YouTube Data API to retrieve updated video statistics and support analysis of more recent data.

3. Machine Learning Integration: Develop machine-learning models to estimate video popularity, predict future views, or estimate engagement levels using suitable historical features.

4. Enhanced Dashboard: Improve the Gradio and Plotly dashboard with additional interactive charts, sorting options, detailed summaries, and user-friendly controls.

5. Category Name Mapping: Map category IDs to their corresponding category names to make the analytical results easier to understand.

6. Data Validation: Investigate unusual records, extreme engagement rates, and inconsistencies to improve the reliability of the analysis.

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APPENDIX

Appendix A: Source Code and Implementation

The source implementation is maintained in the Google Colab notebook Pyspark_project.ipynb. The notebook contains the main stages of the YouTube Video Statistics Analysis project, including SparkSession creation, CSV loading, data inspection, data cleaning, numerical type conversion, video ranking, channel and category aggregation, engagement-rate calculation, and visualization preparation.

The following code excerpt presents the core analytical operations used in the project.

A.1 SparkSession Creation and Dataset Loading

```
None
from pyspark.sql import SparkSessionspark = SparkSession.builder \ .appName("YouTube Video Statistics Analysis") \ .getOrCreate()df = spark.read.csv( "INvideos.csv", header=True, inferSchema=True)
```

A.2 Engagement-Rate Calculation

```
None
from pyspark.sql.functions import col, whenyoutube_df = youtube_df.withColumn( "engagement_rate", when( col("views") > 0, ((col("likes") + col("comment_count")) / col("views")) * 100 ).otherwise(0))
```

A.3 Channel-wise Aggregation

```
None
from pyspark.sql.functions import sum, colchannel_analysis = youtube_df.groupBy("channel_title").agg( sum("views").alias("total_views"), sum("likes").alias("total_likes"), sum("comment_count").alias("total_comments")).orderBy(col("total_views").desc())
```

A.4 Category-wise Aggregation

```
None
category_analysis = youtube_df.groupBy("category_id").agg( sum("views").alias("total_views"), sum("likes").alias("total_likes"), sum("comment_count").alias("total_comments")).orderBy(col("total_views").desc())
```

Appendix B: Dataset Description

The project uses the INvideos.csv dataset containing YouTube trending video statistics.

- Original Pandas inspection: 37,352 rows and 16 columns.
- Spark records before duplicate removal: 38,528 rows.
- Records after duplicate removal: 33,320 rows.
- Final cleaned analysis records: 33,088 rows.

Important dataset attributes include video_id, title, channel_title, category_id, publish_time, views, likes, dislikes, and comment_count. Additional columns contain information such as tags, descriptions, thumbnail links, and comment or rating status.

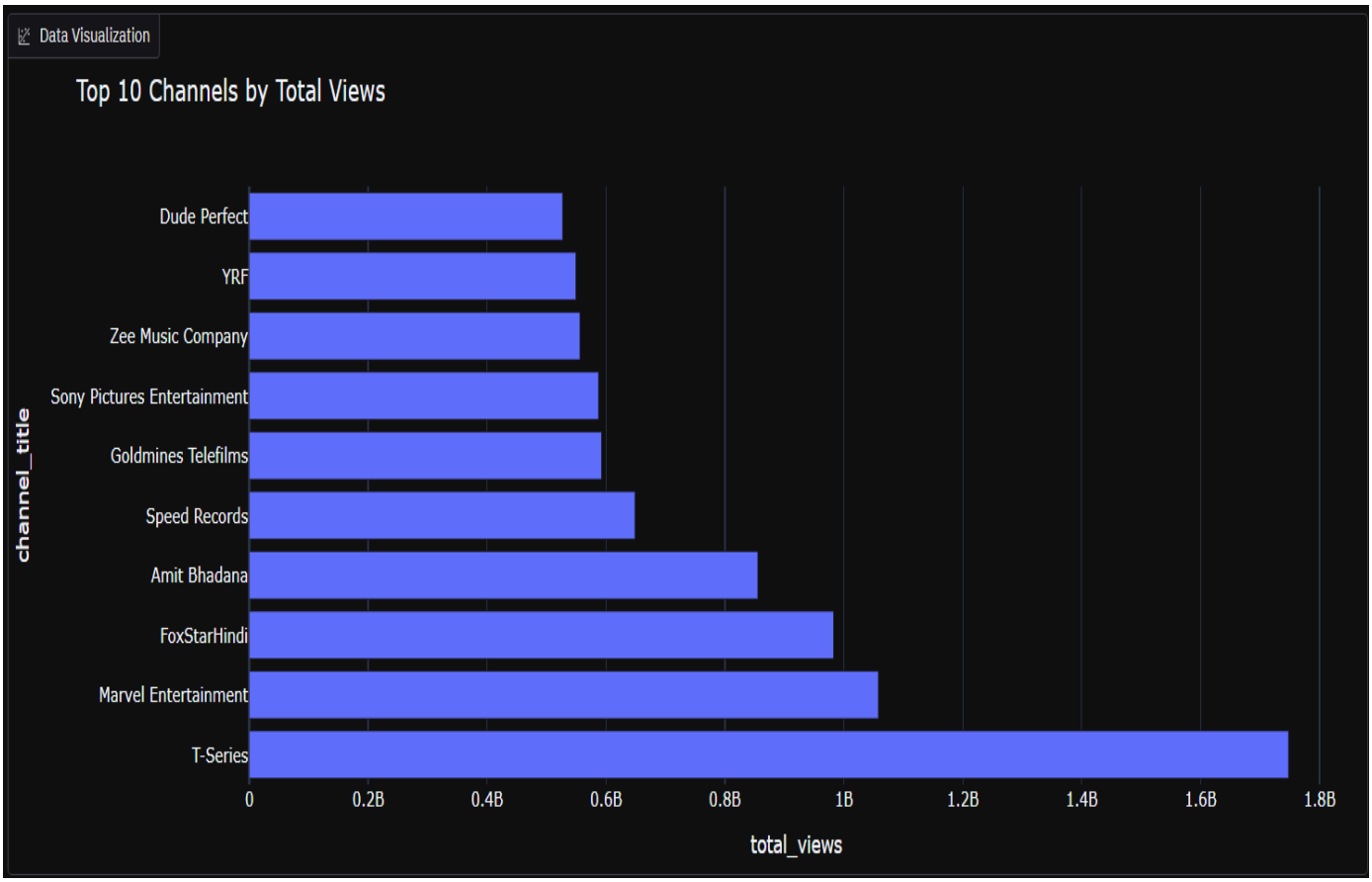
Appendix C: Recommended Screenshots

The following screenshots from the executed notebook can be included as supporting evidence for the implementation and results:

1. Dataset schema and initial inspection.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	video_id	trending_d	title	channel_title	category_id	publish_time	tags	views	likes	dislikes	comment_	thumbnail	comments	ratings_disabled	video_error_or_removed	description		
2	kzwfHumJ	17.14.11	Sharry Ma	Lokdhun Punjabi	1	2017-11-12T1	sharry mar	1096327	33966	798	882	https://i.yt	FALSE	FALSE	FALSE	Presenting Sharry Mann latest Pu		
3	zUZ1z7Fw	17.14.11	à¤ªà¥€à¤°	HJ NEWS	25	2017-11-13T0	à¤ªà¥€à¤°	590101	735	904	0	https://i.yt	TRUE	FALSE	FALSE	à¤ªà¥€à¤°à¤¿à¤¯à¤¡ à¤•à¤		
4	10L1hZ9qz	17.14.11	Stylish Star	TFPC	24	2017-11-12T1	Stylish Star	473988	2011	243	149	https://i.yt	FALSE	FALSE	FALSE	Watch Stylish Star Allu Arjun @ C		
5	N1vE8iiEgt	17.14.11	Eruma Saa	Eruma Saani	23	2017-11-12T0	Eruma Saa	1242680	70353	1624	2684	https://i.yt	FALSE	FALSE	FALSE	This video showcases the differe		
6	kJzGH0PV(	17.14.11	why Sama	Filmylooks	24	2017-11-13T0	Filmylooks	464015	492	293	66	https://i.yt	FALSE	FALSE	FALSE	why Samantha became EMOTIO		
7	il_pSa5l98	17.14.11	MCA (Midd	Dil Raju	24	2017-11-10T0	Nenu Loca	6106669	98612	4185	4763	https://i.yt	FALSE	FALSE	FALSE	Watch MCA- Middle Class Abbay		
8	7MxiQ4v0	17.14.11	Daang (Fu	Speed Records	10	2017-11-11T1	punjabi so	5718766	127477	7134	8063	https://i.yt	FALSE	FALSE	FALSE	Song - Daang\nSinger - Mankirt A		
9	c64I9HNpi	17.14.11	Padmavati	T-Series	10	2017-11-11T0	Ek Dil Ek Ja	10588371	132738	8812	10847	https://i.yt	FALSE	FALSE	FALSE	Presenting the song 'Ek Dil Ek Jaa		
10	KObFEYCa	17.14.11	Chiranjeev	Top Telugu Media	24	2017-11-13T0	Chiranjeev	118223	520	53	23	https://i.yt	FALSE	FALSE	FALSE	Chiranjeevi in Naga Chaitanya - S		

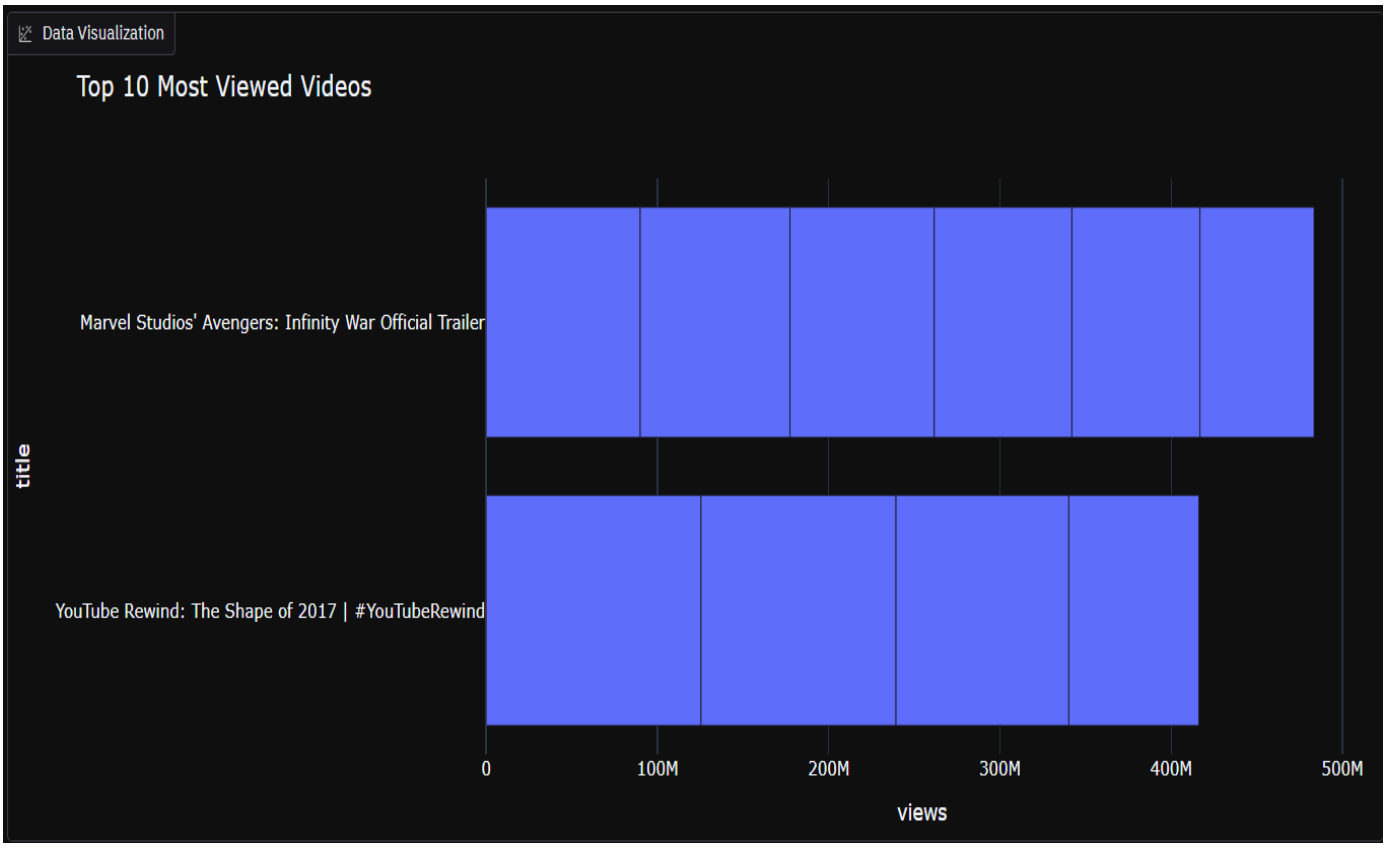
2. Top 10 channels by total views.



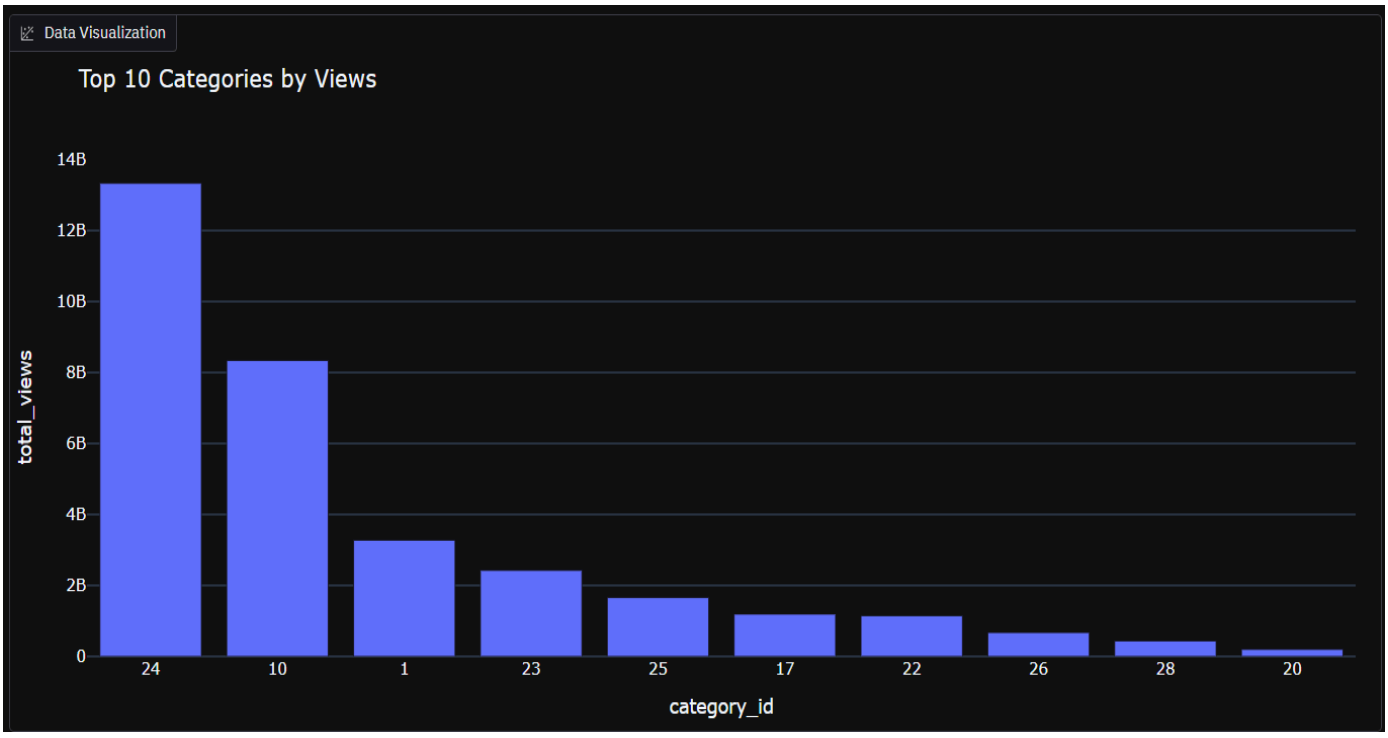
Analysis Results

Results Table

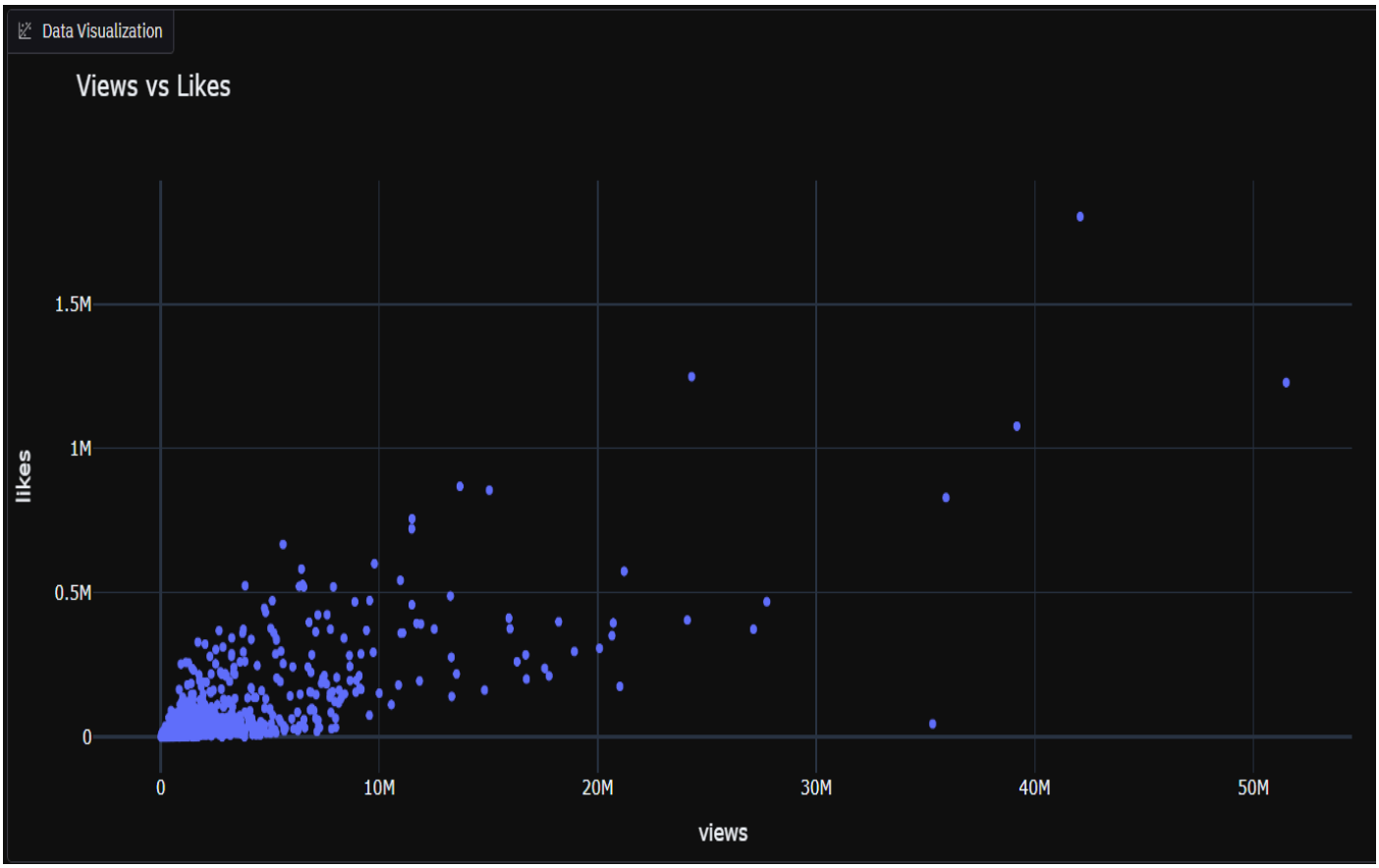
title	channel_title	views	likes	comment_count
YouTube Rewind: The Shape of 2017 | #YouTubeRewind	YouTube Spotlight	125432237	2912710	807558
YouTube Rewind: The Shape of 2017 | #YouTubeRewind	YouTube Spotlight	113876217	2811216	787174
YouTube Rewind: The Shape of 2017 | #YouTubeRewind	YouTube Spotlight	100911567	2656672	682890
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	89930713	2606663	347982
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	87449453	2584674	341571
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	84281319	2555411	339708
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	80360459	2513102	335920
YouTube Rewind: The Shape of 2017 | #YouTubeRewind	YouTube Spotlight	75969469	2251815	827755
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	74789251	2444956	330710
Marvel Studios' Avengers: Infinity War Official Trailer	Marvel Entertainment	66637636	2331352	316185



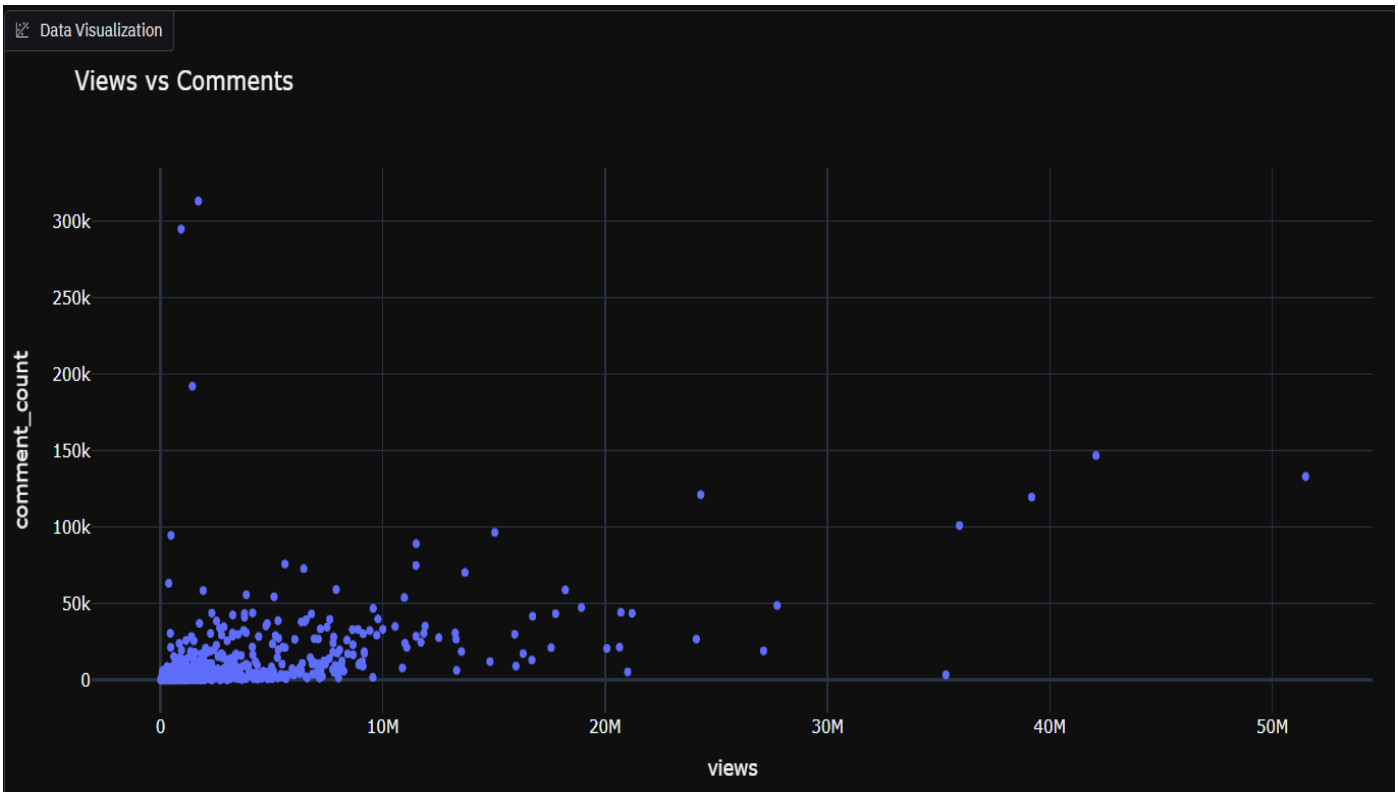
3. Top 10 categories by total views.



4. Scatter plot of views versus likes.



5. Scatter plot of views versus comments.



6. Top 10 records by engagement rate.

